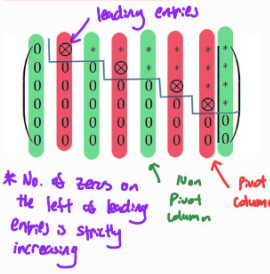
Elementary Row Operations

- Multiplying a row by a nonzero constant;
- Interchanging two rows;
- Adding a multiple of a row to another row.

Row Echelon Form



* RREF, ⊗ = 1

Special Types of Matrices

Square matrix	number of rows and columns are the same	$\begin{pmatrix}0&3&5\\1&-2&3\\4&0&8\end{pmatrix}$
Diagonal matrix	square matrix with all non-diagonal entries being zero	$\begin{pmatrix}1&0&0\\0&3&0\\0&0&2\end{pmatrix}$
Scalar matrix	diagonal matrix with all diagonal entries the same	$\begin{pmatrix}3&0&0\\0&3&0\\0&0&3\end{pmatrix}$
Identity matrix I_n	diagonal matrix with all diagonal entries being 1	$\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$
Zero matrix $0_{m\times n}$	all entries equal to zero	$\begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix}$
Symmetric matrix	k^{th} row "equal" k^{th} column for all k	$\begin{pmatrix}1&-1&0\\-1&3&2\\0&2&2\end{pmatrix}$
Upper triangular matrix	all entries below diagonal are zero	$\begin{pmatrix}1&2&2\\0&3&3\\0&0&2\end{pmatrix}$
Lower triangular matrix	all entries above diagonal are zero	$\begin{pmatrix}1&0&0\\2&3&0\\2&3&2\end{pmatrix}$

Matrix Operations

Matrix Addition	$A+B$	addition of corresponding entries of A and B	$\begin{pmatrix}0&3\\1&-2\end{pmatrix}+\begin{pmatrix}2&7\\8&2\end{pmatrix}=\begin{pmatrix}2&10\\9&0\end{pmatrix}$
Matrix subtraction	$A-B$	subtraction of corresponding entries of A and B	$\begin{pmatrix}0&3\\1&-2\end{pmatrix}-\begin{pmatrix}2&7\\8&2\end{pmatrix}=\begin{pmatrix}-2&-4\\-7&-4\end{pmatrix}$
Scalar multiplication	cA	multiply every entry of A by scalar c	$4\begin{pmatrix}0&3\\1&-2\end{pmatrix}=\begin{pmatrix}0&12\\4&-8\end{pmatrix}$
Negative of matrix	$-A$	attach negative sign to every entry of A	$-\begin{pmatrix}0&3\\1&-2\end{pmatrix}=\begin{pmatrix}0&-3\\-1&2\end{pmatrix}$

Matrix Multiplication Properties

1. Matrix multiplication is not commutative: $AB \neq BA$
2. $AB = 0$ does not imply $A = 0$ or $B = 0$.
3. $A(BC) = (AB)C$ — Associative property
4. $A(B_1 + B_2) = AB_1 + AB_2$ — (Left) Distributive property
5. $(C_1 + C_2)A = C_1A + C_2A$ — (Right) Distributive property
6. $c(AB) = (cA)B = A(cB)$ — c is any scalar (real number)

Matrix Transpose

$$A = \begin{pmatrix}1&2&3&4\\5&6&7&8\end{pmatrix} \qquad A^T = \begin{pmatrix}1&5\\2&6\\3&7\\4&8\end{pmatrix}$$

If A is symmetric, then $A = A^T$.

Let A be an $m \times n$ matrix.

1. $(A^T)^T = A$
2. If B is an $m \times n$ matrix, then $(A + B)^T = A^T + B^T$.
3. If a is a scalar, then $(aA)^T = aA^T$.
4. If B is an $n \times p$ matrix, then $(AB)^T = B^TA^T$.

Matrix Inverse I

Let A be an n x n square matrix.

$AA^{-1} = I$ and $A^{-1}A = I$.

Inverse of 2x2 Matrix

If $A = \begin{pmatrix}a&b\\c&d\end{pmatrix}$,

$$A^{-1} = \begin{pmatrix}\frac{d}{ad-bc}&\frac{-b}{ad-bc}\\\frac{-c}{ad-bc}&\frac{a}{ad-bc}\end{pmatrix}$$

If $ad - bc = 0$, A is singular

Matrix Inverse II

If A is an $n \times n$ matrix:

$$(A|I_n) \xrightarrow{G.J.E} (I_n|A^{-1})$$

Singularity & REF

(i) If the reduced row-echelon form of A is an identity matrix, then A is non-singular.

(ii) If any row-echelon form of A has a zero row, then A is singular.

Matrix Operation on Non-Singular Matrix

A and B are same size & non-singular

Matrix	non-singular?	Inverse
aA	yes	$(aA)^{-1} = (1/a)A^{-1}$
A^T	yes	$(A^T)^{-1} = (A^{-1})^T$
A^{-1}	yes	$(A^{-1})^{-1} = A$
AB	yes	$(AB)^{-1} = B^{-1}A^{-1}$

$$(A_1A_2...A_k)^{-1} = (A_k)^{-1}...(A_2)^{-1}(A_1)^{-1}$$

Singularity & Solutions of Linear Systems

If A is non-singular,

i. The linear system $Ax = b$ has exactly one solution given by $x = A^{-1}b$.

ii. The homogeneous system $Ax = 0$ has only the trivial solution $x = 0$.

If A is singular,

i. The linear system $Ax = b$ has either no solution or infinitely many solutions.

ii. The homogeneous system $Ax = 0$ has both the trivial solution and non-trivial solutions.

Solutions of linear systems

(i) When every column of the row echelon form is a pivot column, except the last column, the system will have exactly one solution.

(ii) When there is at least one non-pivot column in the row echelon form other than the last column, the system will have infinitely many solutions.

(iii) When the very last column of the row echelon form is a pivot column, the system will have no solution.

Matrix Multiplication

$$\begin{matrix}R_1\\R_2\\R_3\end{matrix}\begin{pmatrix}a&b\\c&d\\e&f\end{pmatrix} \times \begin{matrix}C_1&C_2&C_3\end{matrix}\begin{pmatrix}g&b&c\\d&e&f\end{pmatrix}$$

$$= \begin{matrix}R_1\\R_2\\R_3\end{matrix}\begin{matrix}C_1&C_2&C_3\\\end{matrix}\begin{pmatrix}a'+bd&ab+be&ac+bf\\ac+d^2&bc+ed&c^2+df\\ae+df&be+ef&ce+f^2\end{pmatrix}$$

General Solution of non-homogeneous System A

(**) general solution of (A) = general solution of (B) + a particular solution of (A)

The general solution of the non-homogeneous system (A) is the sum of the general solution of the associated homogeneous system (B) and a particular solution of (A).

Non-Singular Matrices

n x n square matrix A

A non-singular matrix	A singular matrix
A has an inverse A^{-1}	A does not have an inverse A^{-1}
row echelon form of A does not have zero row	row echelon form of A has a zero row
reduced row echelon form of A is the identity matrix	reduced row echelon form of A is not the identity matrix
homogeneous system $Ax = 0$ has only trivial solution	homogeneous system $Ax = 0$ has non-trivial solutions in addition to the trivial solution
non-homogeneous system $Ax = b$ has exactly one solution	non-homogeneous system $Ax = b$ has either no solution or infinitely many solutions
det $A \neq 0$	det $A = 0$
Full rank means rank $A = n$	A is full rank / A is not full rank / Not full rank means rank $A < n$

Determinant of 2x2 matrix

$$\det(A) = \begin{vmatrix}a&b\\c&d\end{vmatrix} = ad - bc$$

3x3 matrix

$$\begin{vmatrix}a&b&c\\d&e&f\\g&h&i\end{vmatrix} = a\begin{vmatrix}e&f\\h&i\end{vmatrix} - b\begin{vmatrix}d&f\\g&i\end{vmatrix} + c\begin{vmatrix}d&e\\g&i\end{vmatrix}$$

Det of n x n matrix

n x n determinants is computed in terms of (n-1) x (n-1) determinants

Determinant & Singularity

- if $\det(A) \neq 0$, then A is non-singular;
- if $\det(A) = 0$, then A is singular.

Triangular / Diagonal Matrix

If A is a triangular matrix or diagonal matrix, then the determinant of A is equal to the product of the diagonal entries of A.

$$\begin{vmatrix}-1&3.5&14\\0&5&-26\\0&0&2\end{vmatrix} = (-1)\times5\times2 = -10 \qquad \begin{vmatrix}-2&0&0\\99&0&0\\10&-4.5&10\end{vmatrix} = (-2)\times0\times10 = 0 \qquad \begin{vmatrix}3&0&0\\0&4&0\\0&0&2\end{vmatrix} = 3\times4\times2 = 24$$

Det w/ two identical Rows/Columns

The determinant of a square matrix with two identical rows is zero.
The determinant of a square matrix with two identical columns is zero.

$$\begin{vmatrix}4&-2\\4&-2\end{vmatrix} \qquad \begin{vmatrix}1&2&4\\-1&10&4\\1&2&4\end{vmatrix} \qquad \begin{vmatrix}1&0&0&1\\-1&-3&-3&9\\2&4&4&0\\0&-2&-2&-1\end{vmatrix}$$

Det & Elementary Row Operations

E.R.O	Determinant
$A \xrightarrow{kR_i} B$	$\det(B) = k\det(A)$
$A \xrightarrow{R_i \leftrightarrow R_j} B$	$\det(B) = -\det(A)$
$A \xrightarrow{R_i + kR_j} B$	$\det(B) = \det(A)$

Det & Matrix Operation

Let A and B be n x n square matrices, and c be a scalar.

1. $\det(cA) = c^n\det(A)$
2. $\det(AB) = \det(A)\det(B)$
3. $\det(A^T) = \det(A)$
4. if A is non-singular, then $\det(A^{-1}) = \frac{1}{\det(A)}$
5. $\det(A + B) \neq \det(A) + \det(B)$

Rank of a matrix

Let A be an m x n matrix and R be a row echelon form of A.

$A \xrightarrow{G.E.} R$ (REF)

The rank of A is the number of nonzero rows of R.

rank(A) = the number of leading entries in R
= the number of pivot columns in R

$$A = \begin{pmatrix}1&2&2&1\\3&6&6&3\\4&9&9&5\\-2&-1&-1&1\\5&8&9&4\\4&2&7&3\end{pmatrix} \xrightarrow{\text{Gaussian Elimination}} \begin{pmatrix}\boxed{1}&2&2&1\\0&\boxed{1}&1&1\\0&0&0&\boxed{1}\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

Non-zero rows

rank(A) = 3

Pivot Columns

Full Rank

$\text{rank}(A) \leq \min\{m, n\}$.

the smaller of the two numbers m and n

Rank & Determinant

A is full rank $\Leftrightarrow$ $\det(A) \neq 0$

RREF of A is I $\Leftrightarrow$ A is non-singular

Three products A, B, C

Production cost per item

	A	B	C
Raw materials	0.10	0.30	0.15
Labor	0.30	0.40	0.25
Overhead & Misc.	0.10	0.20	0.13

$$\begin{matrix}0.10&0.30&0.15\\0.30&0.40&0.25\\0.10&0.20&0.13\end{matrix}$$ Easier to manipulate the data

Production cost matrix

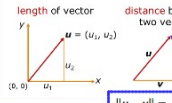
Vector Operations

Notation	Algebraic	Geometric
Vector addition $u+v$	$(u_1+v_1, u_2+v_2, ..., u_n+v_n)$	
Vector subtraction $u-v$	$(u_1-v_1, u_2-v_2, ..., u_n-v_n)$	
Negative $-u$	$(-u_1, -u_2, ..., -u_n)$	
Scalar multiplication cu	$(cu_1, cu_2, ..., cu_n)$	

Dot Product

$$u \cdot v = u_1v_1 + u_2v_2 + \ldots + u_nv_n.$$

Length, Distance and Angles in R^2



$\|u\| = \sqrt{u_1^2 + u_2^2}$. $\|u - v\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$ $\cos\theta = \frac{u_1v_1 + u_2v_2}{\|u\|\,\|v\|}$

Length, Distance & Angles in R^n

$\|u\| = \sqrt{u \cdot u}$, $\|u - v\| = \sqrt{(u - v) \cdot (u - v)}$, $\theta = \cos^{-1}\left(\frac{u \cdot v}{\|u\|\,\|v\|}\right)$.

For a general n-vector, the length is also known as the norm.

Orthogonal Vectors

Let u and v be two vectors in R^n. If $u \cdot v = 0$, we say u and v are orthogonal.

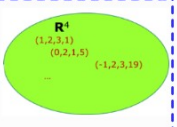
In this case, the angle between u and v:

$\theta = \cos^{-1}\left(\frac{u \cdot v}{\|u\|\,\|v\|}\right) = \cos^{-1}(0) = \frac{\pi}{2}$. $u \cdot v = 0$

n Space

For a fix number n, the set or collection of all the n-vectors is called the n-space, and is represented by the notation R^n.

R^2: xy plane, R^3: xyz plane



The following are different ways to say the same thing:

- $u \in R^n$
- u is an n-vector
- $u = (u_1, u_2, ..., u_n)$ for some real numbers $u_1, u_2, ..., u_n$.

In general, if a linear system has n variables, then its solutions are n-vectors, and the solution set is a subset of R^n.

Linear Combination

v is a linear combination of u_1, u_2, u_3 — can find a,b,c → $v = au_1 + bu_2 + cu_3$

v is not a linear combination of u_1, u_2, u_3 — cannot find a,b,c

linear system has no solution

$(3, 3, 3, 0) = a(2, 1, 3, 1) + b(1, -1, 2, 2) + c(3, 0, 5, 1)$

$$\begin{cases}2a+b+3c=3\\a-b=3\\3a+2b+5c=3\\a+2b+c=0\end{cases} \rightarrow \begin{pmatrix}2&1&3&3\\1&-1&0&3\\3&2&5&3\\1&2&1&0\end{pmatrix} \xrightarrow{G.E} \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}$$

linear system has solution

$v = (3, 3, 4, 0)$

$(3, 3, 4, 0) = a(2, 1, 3, 1) + b(1, -1, 2, 2) + c(3, 0, 5, 1)$

$$\begin{cases}2a+b+3c=3\\a-b=3\\3a+2b+5c=4\\a+2b+c=0\end{cases} \rightarrow \begin{pmatrix}2&1&3&3\\1&-1&0&3\\3&2&5&4\\1&2&1&0\end{pmatrix} \xrightarrow{G.E} \begin{pmatrix}1&0&0&2\\0&1&0&-1\\0&0&1&0\\0&0&0&0\end{pmatrix}$$

Linear Systems

$$\begin{cases}2a+b+3c=3\\a-b=3\\3a+2b+5c=4\\a+2b+c=0\end{cases}$$ standard form

$$a\begin{pmatrix}2\\1\\3\\1\end{pmatrix}+b\begin{pmatrix}1\\-1\\2\\2\end{pmatrix}+c\begin{pmatrix}3\\0\\5\\1\end{pmatrix}=\begin{pmatrix}3\\3\\4\\0\end{pmatrix}$$ vector equation form

$$\begin{pmatrix}2&1&3\\1&-1&0\\3&2&5\\1&2&1\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}3\\3\\4\\0\end{pmatrix}$$ matrix equation form

Row Space

$$A = \begin{pmatrix}2&-1&0\\1&-1&3\\-5&1&0\\1&0&1\end{pmatrix}$$ rows of A

$r_1 = (2, -1, 0)$
$r_2 = (1, -1, 3)$
$r_3 = (-5, 1, 0)$
$r_4 = (1, 0, 1)$

row space of $A = \text{span}\{r_1, r_2, r_3, r_4\} \neq \{r_1, r_2, r_3, r_4\}$.

Column Space

$$A = \begin{pmatrix}2&-1&0\\1&-1&3\\-5&1&0\\1&0&1\end{pmatrix}$$ columns of A

$$c_1 = \begin{pmatrix}2\\1\\-5\\1\end{pmatrix} \quad c_2 = \begin{pmatrix}-1\\-1\\1\\0\end{pmatrix} \quad c_3 = \begin{pmatrix}0\\3\\0\\1\end{pmatrix}$$

column space of $A = \text{span}\{c_1, c_2, c_3\} \neq \{c_1, c_2, c_3\}$

Null Space

$$A = \begin{pmatrix}1&-2&3\\2&-4&6\\3&-6&9\end{pmatrix}$$

$$\begin{pmatrix}1&-2&3\\2&-4&6\\3&-6&9\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \begin{matrix}x-2y+3z=0\\2x-4y+6z=0\\3x-6y+9z=0\end{matrix}$$

general solution $\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2s-3t\\s\\t\end{pmatrix} = s\begin{pmatrix}2\\1\\0\end{pmatrix} + t\begin{pmatrix}-3\\0\\1\end{pmatrix}$

the nullspace of A = span $\left\{\begin{pmatrix}2\\1\\0\end{pmatrix}, \begin{pmatrix}-3\\0\\1\end{pmatrix}\right\}$

Subspaces

given an m x n matrix A:

The row space of A is a subspace of R^n;

The column space of A is a subspace of R^m;

The nullspace of A is a subspace of R^n.

Closure Properties

A subspace V satisfies the closure properties:

(i) for all $u, v \in V$, we must have $u + v \in V$.

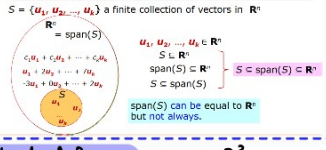
(ii) for all $u \in V$ and $c \in R$, we must have $cu \in V$.

Span

span(S) is the set of all linear combinations of $u_1, u_2, ..., u_k$

Can also be written: span$\{u_1, u_2, ..., u_k\}$

$S = \{u_1, u_2, ..., u_k\}$ a finite collection of vectors in R^n



$u_1, u_2, ..., u_k \in R^n$

$S \subseteq R^n$

span(S) $\subseteq R^n$

$S \subseteq$ span(S)

$S \subseteq \text{span}(S) \subseteq R^n$

span(S) can be equal to R^n but not always.

Basis for Column Space

$$\text{rref}(A) = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\\0&0&0\end{pmatrix} \qquad A = \begin{pmatrix}2&-1&0\\1&-1&3\\-5&1&0\\1&0&1\end{pmatrix}$$

Pivot Columns

Basis for column space = $\left\{\begin{pmatrix}2\\1\\-5\\1\end{pmatrix}, \begin{pmatrix}-1\\-1\\1\\0\end{pmatrix}, \begin{pmatrix}0\\3\\0\\1\end{pmatrix}\right\}$

the columns of A corresponding to pivot columns of R → a basis for the column space of A.

dim(column space of A) = rank(A).

Basis for Row Space

$$\text{rref}(A) = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\\0&0&0\end{pmatrix}$$

A basis for row space = $\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$

the non-zero rows in any r.e.f. R of a matrix A → a basis for the row space of A.

dim(row space of A) = rank(A).

dim(row space of A) = dim(column space of A)

Linear Independance

Let $u_1, u_2, ..., u_k$ be a set of vectors in R^n.
Set up the vector equation
$c_1u_1 + c_2u_2 + \cdots + c_ku_k = 0$.

Case I: The system has only the trivial solution, i.e. the only possible scalars are $c_1 = 0$, $c_2 = 0$, ..., $c_k = 0$

In this case, we say the set of vectors $u_1, u_2, ..., u_k$ are linearly independent

Case II: The system has non-trivial solutions, i.e. there are scalars c_1, c_2, ..., c_k that are not all zero.

In this case, we say the set $u_1, u_2, ..., u_k$ are linearly dependent.

$$\begin{pmatrix}1&0&1&0\\0&2&-1&0\\0&1&1&0\\1&0&1&0\end{pmatrix} \xrightarrow{G.E.} \begin{pmatrix}1&0&1&0\\0&2&-1&0\\0&0&1&0\\0&0&0&0\end{pmatrix}$$ - Only trivial solutions - Linearly Independent

$$\begin{pmatrix}1&5&3&0\\-2&6&2&0\\3&-1&1&0\end{pmatrix} \xrightarrow{G.E.} \begin{pmatrix}1&5&3&0\\0&16&8&0\\0&0&0&0\end{pmatrix}$$ - Has non-trivial solutions - Linearly dependent

Linear Independance & Determinant

- If $\det(A) = 0$, linearly dependent
- If $\det(A) \neq 0$, linearly independent.

Solution space of $Ax = 0$ → subspace of R^n

Row space of a matrix A → subspace of R^n

Column space of a matrix A → subspace of R^m

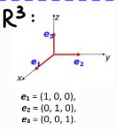
Nullspace of a matrix A → subspace of R^n

Standard Basis

In general, for any positive integer n, the standard basis vectors for R^n:

$b_1 = (1, 0, ..., 0)$, $b_2 = (0, 1, ..., 0)$, ..., $b_n = (0, 0, ..., 1)$

R^3:



$e_1 = (1, 0, 0)$,
$e_2 = (0, 1, 0)$,
$e_3 = (0, 0, 1)$.

Basis for Solution Space

Nullspace of **A** = solution space of **Ax = 0**
So it is a subspace of $\mathbb{R}^n$

$$\begin{cases} 2v + 2w - x + z = 0 \\ -v - w + 2x - 3y + z = 0 \\ x + y + z = 0 \\ v + w - 2x - z = 0 \end{cases}$$

After performing Gaussian elimination, the general solution is given on the left side below:

$$\begin{pmatrix} v \\ w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -s-t \\ s \\ -t \\ 0 \\ t \end{pmatrix} = s\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \qquad u_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

$\{u_1, u_2\}$ is a basis for the solution space of the given system, and dim(solution space) = 2.

dim(solution space) = number of parameters in the general solution.

Dimension Theorem of Matrix

Nullspace of matrix **A**	Solution space of system **Ax = 0**
Basis for nullspace	Basis for solution space
Dimension of nullspace	Dimension of solution space
Nullity of A	# parameters in general solution
	# non-pivot columns of RREF

If **A** is a matrix with n columns, then rank(**A**) + nullity(**A**) = n.

rank(**A**) = dim(column space of **A**)
nullity(**A**) = dim(nullspace of **A**).

Least Square Solutions

When a linear system **Ax = b** is inconsistent

A least squares solution $\mathbf{x}_0$ of **Ax = b** :

- $\mathbf{x}_0$ is the best approximation to a solution of **Ax = b**

$$\underset{\text{closest}}{\mathbf{Ax}_0} \approx \mathbf{b} \qquad \underset{\text{smallest}}{||\mathbf{Ax}_0 - \mathbf{b}||}$$

How to find the least squares solution $\mathbf{x}_0$ of **Ax = b**?

- Multiply $\mathbf{A}^T$ on both sides of **Ax = b**
- Solve the new system $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$

Projection on Subspaces

If $\mathbf{x}_0$ is the least squares solution of **Ax = b**, $\mathbf{Ax}_0$ = projection of **b** onto column space of **A**

Given a subspace U of $\mathbb{R}^n$ and a vector **b** not in U

Find a vector **p** in U "nearest" to **b**

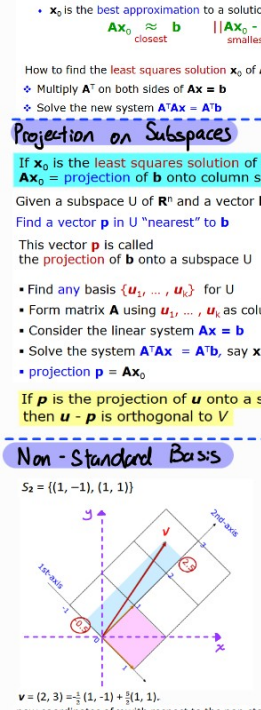
This vector **p** is called the projection of **b** onto a subspace U

- Find any basis $\{u_1, \dots, u_k\}$ for U
- Form matrix **A** using $u_1, \dots, u_k$ as column vectors
- Consider the linear system **Ax = b**
- Solve the system $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$, say $\mathbf{x}_0$
- projection $\mathbf{p} = \mathbf{Ax}_0$

If **p** is the projection of **u** onto a subspace V, then **u - p** is orthogonal to V

Non-Standard Basis

$S_2 = \{(1, -1), (1, 1)\}$



$\mathbf{v} = (2, 3) = \frac{1}{2}(1, -1) + \frac{5}{2}(1, 1)$.
new coordinates of **v** with respect to the non-standard basis S_2:
$(\mathbf{v})_{S_2} = (-\frac{1}{2}, \frac{5}{2})$.

Coordinate Vectors

Let $S = \{u_1, u_2, \dots, u_k\}$ be a basis for a vector space V and **v** a fixed vector in V. **v** can be written as a linear combination of this basis

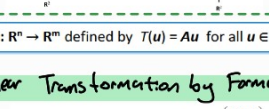
$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k$$

coordinate vector of **v** relative to the basis S:

$$(\mathbf{v})_S = (c_1, c_2, \dots, c_k)$$

Linear Transformation

$$\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \longrightarrow \mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \longrightarrow \mathbf{Au} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



$T: \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{u}) = \mathbf{Au}$ for all $\mathbf{u} \in \mathbb{R}^n$.

Linear Transformation by Formula

$T: \mathbb{R}^2 \to \mathbb{R}^3$ such that $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ 2x \\ -3y \end{pmatrix}$

$$= x\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 0 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

Properties of Linear Transformation

1. $T(\mathbf{0}) = \mathbf{0}$
2. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
3. $T(c\mathbf{u}) = cT(\mathbf{u})$
4. $T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k) = c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + \cdots + c_kT(\mathbf{u}_k)$

Images of Standard Basis

Let $T: \mathbb{R}^n \to \mathbb{R}^m$ be any linear transformation with the standard matrix given by

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

the image $T(\mathbf{e}_k)$ = the kth column of **A** and and the matrix $\mathbf{A} = (T(\mathbf{e}_1)\ T(\mathbf{e}_2)\ \cdots\ T(\mathbf{e}_n))$.

Stacking Method

$$T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \qquad T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$\mathbf{A}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \text{ and } \mathbf{A}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} \quad (*)$$

$$\mathbf{A}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 2 & 6 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 4 & 0 \\ 2 & 6 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 2 \\ 4 & -2 \end{pmatrix}.$$

Eigenvalues & Eigen Vectors

Square matrix **A** → eigen vector **v** → **Av** multiple of **v** → scalar → eigenvalue associated with the eigenvector

When input and output vectors are scalar multiples

- If an eigenvector **u** is given, multiply it by the matrix: $\mathbf{Au} = \lambda\mathbf{u}$
- Solve characteristic equation $\det(\lambda\mathbf{I} - \mathbf{A}) = 0$
- If the matrix is triangular, take the diagonal entries
- If λ is an eigenvalue of **A**, then
  - λ is an eigenvalue of $\mathbf{A}^T$
  - λ^n is an eigenvalue of $\mathbf{A}^n$
  - λ^{-1} is an eigenvalue of $\mathbf{A}^{-1}$ (when **A** is non-singular)

If 0 is an eigenvalue of **A**, then **A** is singular. Otherwise, **A** is non-singular.

0 is an eigenvalue of **A**
0 is a root of the characteristic polynomial $\det(\lambda\mathbf{I} - \mathbf{A})$
$\det(0\mathbf{I} - \mathbf{A}) = 0$
$\det(-\mathbf{A}) = 0$
$\det(\mathbf{A}) = 0$
A is singular

Useful Observations for Matrix Multiplication

Observation 1: Matrix multiplication of two matrices **A** and **B**.

If we write matrix **B** in terms of its columns $\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n$. Then

$$\mathbf{AB} = \mathbf{A}(\mathbf{b}_1\ \mathbf{b}_2\ \cdots\ \mathbf{b}_n) = (\mathbf{Ab}_1\ \mathbf{Ab}_2\ \cdots\ \mathbf{Ab}_n)$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 2 & 5 & 8 \\ 2 & 4 & 6 \end{pmatrix}$$
$$\mathbf{A} \qquad \mathbf{b}_1\ \mathbf{b}_2\ \mathbf{b}_3 \qquad \mathbf{Ab}_1\ \mathbf{Ab}_2\ \mathbf{Ab}_3$$

Observation 2: Matrix multiplication of two matrices **B** and **D**, where **D** is a diagonal matrix.

$$\mathbf{BD} = (\mathbf{b}_1\ \mathbf{b}_2\ \cdots\ \mathbf{b}_n)\,\mathbf{D} = (d_1\mathbf{b}_1\ d_2\mathbf{b}_2\ \cdots\ d_n\mathbf{b}_n)$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{pmatrix}\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 20 \end{pmatrix}$$
$$\mathbf{b}_1\ \mathbf{b}_2\ \mathbf{b}_3 \qquad \mathbf{D} \qquad 2\mathbf{b}_1\ 3\mathbf{b}_2\ 4\mathbf{b}_3$$

Power of Matrix with Eigenvectors

$$\mathbf{A} = \begin{pmatrix} 0.96 & 0.01 \\ 0.04 & 0.99 \end{pmatrix} \qquad \lambda_1 = 1,\ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$
$$\lambda_2 = 0.95,\ \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\{\mathbf{v}_1, \mathbf{v}_2\}$ is the basis for $\mathbb{R}^2$

Express $\mathbf{x}_0$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_2$

$$\mathbf{A}^n\mathbf{x}_0 = \mathbf{A}^n(c_1\mathbf{v}_1 + c_2\mathbf{v}_2)$$
$$= c_1\mathbf{A}^n\mathbf{v}_1 + c_2\mathbf{A}^n\mathbf{v}_2$$
$$= c_1\lambda_1^n\mathbf{v}_1 + c_2\lambda_2^n\mathbf{v}_2$$

First Order System of Differential Equation

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -0.02 & 0.02 \\ 0.02 & -0.02 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

notation for the SDE:

$$\mathbf{y}'(t) = \mathbf{A}\,\mathbf{y}(t), \text{ or simply } \mathbf{y}' = \mathbf{A}\,\mathbf{y}.$$

Eigenspace

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Solve $(\lambda\mathbf{I} - \mathbf{A})\,\mathbf{x} = \mathbf{0}$ for eigenvalues $\lambda = 1$ and -1

$$\begin{pmatrix} \lambda & 0 & -1 & 0 \\ 0 & \lambda & 0 & -1 \\ -1 & 0 & \lambda & 0 \\ 0 & -1 & 0 & \lambda \end{pmatrix}\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$\lambda = 1$

$$\begin{pmatrix} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{\text{G.E.}} \begin{pmatrix} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

w = t, z = s, y = t, x = s

dim E_1 = 2

Basis for Eigenspace E_1

Gen. sol. $\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} s \\ t \\ s \\ t \end{pmatrix} = s\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

Eigenspace for $\lambda = 1$: $E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}\right\}$

Any linear combination is an eigenvector associated to $\lambda = 1$

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Characteristic polynomial $(\lambda - 1)^2(\lambda + 1)^2$
Eigenvalues of **A** are 1 and -1
Eigenspaces of **A** are E_1 and E_{-1}:

$E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\right\}$ dim E_1 = 2

$E_{-1} = \text{span}\left\{\begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}\right\}$ dim E_{-1} = 2

Example

Characteristic polynomial of **B**: $(\lambda - 1)^3(\lambda - 2)^3(\lambda - 4)^2$
Eigenvalues of **B** are 1, 2 and 4 with multiplicity 1, 3, 2 respectively
Eigenspaces of **B** are E_1, E_2 and E_4:
dim E_1 = 1 dim $E_2 \le 3$ dim $E_4 \le 2$

Dimension of Eigenspace

Suppose the characteristic polynomial of a matrix **A** is factorized as follow

$$\det(\lambda\mathbf{I} - \mathbf{A}) = (\lambda - \lambda_1)^{r_1}(\lambda - \lambda_2)^{r_2}\cdots(\lambda - \lambda_k)^{r_k}$$

If **A** is an n x n matrix, then

$$r_1 + r_2 + \dots + r_k = n$$
$$\dim E_{\lambda_i} \le r_i \text{ for all } i$$

Diagonalizable Matrix

An n x n square matrix **A** is called diagonalizable if we can find a non-singular matrix **P** such that $\mathbf{P}^{-1}\mathbf{AP}$ is a diagonal matrix.

$$\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$$

Suppose **A** is an n x n diagonalizable matrix. Use the *n* linearly independent eigenvectors to form the *n* columns of a matrix **P**. This non-singular **P** will diagonalize **A**.

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

eigenvalues $\lambda = 1, -1$

$E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}\right\}$ $E_{-1} = \text{span}\left\{\begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}\right\}$

A is diagonalizable

$$\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D} \Longleftarrow \mathbf{P} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\mathbf{A} = \mathbf{PDP}^{-1}$$

If **A** has *n* linearly independent eigenvectors, then **A** is diagonalizable

Power of Matrix

$$\mathbf{A}^n = \mathbf{PD}\,\mathbf{D}\cdots\mathbf{DP}^{-1} = \mathbf{PD}^n\mathbf{P}^{-1}$$

$$\mathbf{D}^n = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_n \end{pmatrix}^n = \begin{pmatrix} \lambda_1^n & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_n^n \end{pmatrix}$$

Solutions of y' = Ay

If **v** is an eigenvector of an n x n matrix **A** with eigenvalue r :
Then $\mathbf{y} = \mathbf{v}e^{rt}$ is a solution of the SDE $\mathbf{y}' = \mathbf{Ay}$

Suppose **A** has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ with eigenvalues $r_1, r_2, \dots, r_n$.
$\mathbf{v}_1e^{r_1t}, \mathbf{v}_2e^{r_2t}, \dots, \mathbf{v}_ne^{r_nt}$ are n linearly independent solutions
$\{\mathbf{v}_1e^{r_1t}, \mathbf{v}_2e^{r_2t}, \dots, \mathbf{v}_ne^{r_nt}\}$ forms a a fundamental set of solutions
$\mathbf{y} = c_1\mathbf{v}_1e^{r_1t} + c_2\mathbf{v}_2e^{r_2t} + \dots + c_n\mathbf{v}_ne^{r_nt}$ gives the general solution of the SDE

Super Position Principal

If $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are two solutions to an SDE $\mathbf{y}' = \mathbf{Ay}$, then

$$\mathbf{y} = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$$

is also a solution to the system for any scalars c_1 and c_2.

$$\mathbf{y}' = c_1\mathbf{x}_1'(t) + c_2\mathbf{x}_2'(t)$$
$$\mathbf{y}' = c_1\mathbf{Ax}_1(t) + c_2\mathbf{Ax}_2(t)$$
$$\mathbf{y}' = \mathbf{A}(\,c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)\,)$$

Initial Value Problem

SDE $\mathbf{y}' = \mathbf{Ay}$ + initial condition $\mathbf{y}(t_0) = \mathbf{v}$

→ general solution

$\mathbf{y}(t_0) = c_1\mathbf{v}_1e^{r_1t_0} + c_2\mathbf{v}_2e^{r_2t_0} + \dots + c_n\mathbf{v}_ne^{r_nt_0}$

Become a linear system

→ substitute $t = t_0$

Solve for specific values:
$c_1 = a_1, c_2 = a_2, \dots, c_n = a_n$

This solution is unique

Particular solution (unique)
$\mathbf{y}_p = a_1\mathbf{v}_1e^{r_1t} + a_2\mathbf{v}_2e^{r_2t} + \dots + a_n\mathbf{v}_ne^{r_nt}$

Converting ODE to SDE

- Introduce variables y_1 and y_2
  $y_1 = y$
  $y_2 = y'$
- Express y_1' and y_2' in terms of y_1 and y_2

$$ay'' + by' + cy = 0$$

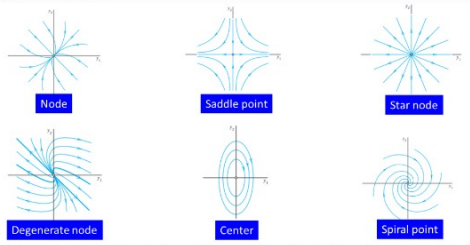
$$\begin{cases} y_1' = y_2 \\ y_2' = -\frac{c}{a}y_1 - \frac{b}{a}y_2 \end{cases}$$

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Eigen Value Criterion, 2x2 matrix A

Eigenvalues r_1 and r_2	Equilibrium Point	Stability
Real & distinct (both positive)	Node	Unstable
Real & distinct (both negative)	Node	Asymptotically stable
Real & repeated (positive) (2 indep. eigenvectors)	Star node	Unstable
Real & repeated (negative) (2 indep. eigenvectors)	Star node	Asymptotically stable
Real & repeated (positive) (1 indep. eigenvector)	Degenerate node	Unstable
Real & repeated (negative) (1 indep. eigenvector)	Degenerate node	Asymptotically stable
Real (opposite sign)	Saddle point	Unstable
Complex (positive real parts)	Spiral point	Unstable
Complex (negative real parts)	Spiral point	Asymptotically stable
Pure imaginary	Center	Neutrally stable

Phase Portrait

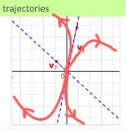


Node (+ve)

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 1$ and $r_2 = 6$ $\quad r_1 < r_2$
Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ for r_1, and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2

real and distinct, both positive
→ equilibrium point: Node, unstable



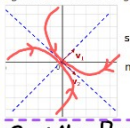
start near origin and parallel to $\mathbf{v}_1$ (smaller eigenvalues r_1)
move away from origin and parallel to $\mathbf{v}_2$ (larger eigenvalues r_2)

Node (-ve)

$\mathbf{y}' = \mathbf{Ay}$

Eigenvalues $r_1 = -3$ and $r_2 = -2$
Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for r_1, and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for r_2

Eigenvalues: real and distinct → Node
Eigenvalues: both negative → Asymptotically stable



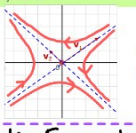
start (far) away from origin and parallel to $\mathbf{v}_1$ (smaller eigenvalues r_1)
move towards origin and parallel to $\mathbf{v}_2$ (larger eigenvalues r_2)

Saddle Point

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = -4$ and $r_2 = 1$ $\quad r_1 < r_2$
Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ for r_1, and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2

real and distinct, opposite sign
→ equilibrium point: saddle point, unstable



Start parallel to $\mathbf{v}_1$ and moving towards origin
End parallel to $\mathbf{v}_2$ and moving away from origin

Complex Eigenvalues

Eigenvalues $r_1 = a + b\,i$ and $r_2 = a - b\,i$ (complex conjugates)

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} c + d\,i \\ e + f\,i \end{pmatrix} = \mathbf{p} + \mathbf{q}\,i$ and $\mathbf{v}_2 = \begin{pmatrix} c - d\,i \\ e - f\,i \end{pmatrix} = \mathbf{p} - \mathbf{q}\,i$

Complex solutions
$\mathbf{x}_1(t) = \mathbf{v}_1e^{r_1t}$
$\mathbf{x}_2(t) = \mathbf{v}_2e^{r_2t}$

Real solutions
$\mathbf{x}_{Re}(t) = e^{\alpha t}(\mathbf{p}\cos bt - \mathbf{q}\sin bt)$
$\mathbf{x}_{Im}(t) = e^{\alpha t}(\mathbf{p}\sin bt + \mathbf{q}\cos bt)$

General solution
$\mathbf{y} = c_1\mathbf{x}_{Re}(t) + c_2\mathbf{x}_{Im}(t)$

Spiral Point

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{matrix} 2y_1 - 4y_2 \\ 4y_1 + 2y_2 \end{matrix} \qquad \mathbf{A} = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$$

- Find eigenvalues and eigenvectors
- Find fundamental set of real solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2 + 4\,i$, $r_2 = 2 - 4\,i$
$a = 2$, $b = 4$
Eigenvector $\mathbf{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$ $\quad \mathbf{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$
$\mathbf{p} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

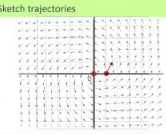
Fundamental set of (real) solutions
$\mathbf{x}_{Re}(t) = e^{at}(\mathbf{p}\cos bt - \mathbf{q}\sin bt) = e^{2t}(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\cos 4t - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\sin 4t)$
$\mathbf{x}_{Im}(t) = e^{at}(\mathbf{p}\sin bt + \mathbf{q}\cos bt) = e^{2t}(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\sin 4t + \begin{pmatrix} 1 \\ 0 \end{pmatrix}\cos 4t)$

General solution
$\mathbf{y} = c_1\mathbf{x}_{Re}(t) + c_2\mathbf{x}_{Im}(t) = e^{2t}\left[c_1\begin{pmatrix} -\sin(4t) \\ \cos(4t) \end{pmatrix} + c_2\begin{pmatrix} \cos(4t) \\ \sin(4t) \end{pmatrix}\right]$

- Find eigenvalues and eigenvectors
- Find fundamental set of real solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2 + 4\,i$, $r_2 = 2 - 4\,i$
Eigenvector $\mathbf{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$ $\quad \mathbf{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$

Complex, real part > 0
→ equilibrium point: spiral point, unstable



Orientation of trajectories
- Test a point, say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Multiply by **A** to get $\begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$
- Tangent vector $\mathbf{y}'$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$, pointing upward

Center

$\begin{cases} y_1' = 2y_1 + 4y_2 \\ y_2' = -2y_1 - 2y_2 \end{cases}$ $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix}$

- Find eigenvalues and eigenvectors
- Find fundamental set of real solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2i$, $r_2 = -2i$
$a = 0$, $b = 2$
Eigenvector $\mathbf{v}_1 = \begin{pmatrix} -1 - i \\ 1 \end{pmatrix}$ $\mathbf{v}_2 = \begin{pmatrix} -1 + i \\ 1 \end{pmatrix}$
$\mathbf{p} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$
→ purely imaginary (real part = 0)
→ equilibrium point: center, neutrally stable

Fundamental set of (real) solutions

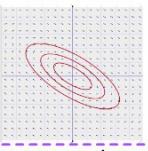
$\mathbf{x}_{Re}(t) = e^{at}(\mathbf{p}\cos bt - \mathbf{q}\sin bt) = \begin{pmatrix} -1 \\ 1 \end{pmatrix}\cos 2t - \begin{pmatrix} -1 \\ 0 \end{pmatrix}\sin 2t$

$\mathbf{x}_{Im}(t) = e^{at}(\mathbf{p}\sin bt + \mathbf{q}\cos bt) = \begin{pmatrix} -1 \\ 1 \end{pmatrix}\sin 2t + \begin{pmatrix} -1 \\ 0 \end{pmatrix}\cos 2t$

What is the orientation of the trajectories?

General solution

$\mathbf{y} = c_1\mathbf{x}_{Re}(t) + c_2\mathbf{x}_{Im}(t) = c_1\begin{pmatrix} -\cos(2t) + \sin(2t) \\ \cos(2t) \end{pmatrix} + c_2\begin{pmatrix} -\sin(2t) - \cos(2t) \\ \sin(2t) \end{pmatrix}$



Orientation of trajectories

- Test a point, say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Multiply by $\mathbf{A}$ to get $\begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$
- Tangent vector $\mathbf{y}'$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$, pointing downward

Repeated Eigenvalues

A (2x2) eigenvalue r	Diagonalizable (scalar matrix)	Not diagonalizable
eigenvectors	2 linearly independent $\mathbf{v}_1$, $\mathbf{v}_2$ (every non-zero vector is eigenvector)	1 linearly independent $\mathbf{v}$
Equilibrium point	Star node	Degenerate node 
Stability	Unstable (r > 0) Asymptotically stable (r < 0)	Unstable (r > 0) Asymptotically stable (r < 0)
General solution	$\mathbf{y} = c_1\begin{pmatrix} e^{rt} \\ 0 \end{pmatrix} + c_2\begin{pmatrix} 0 \\ e^{rt} \end{pmatrix}$	$\mathbf{y} = c_1e^{rt}\mathbf{v} + c_2e^{rt}(t\mathbf{v} + \mathbf{u})$ $\mathbf{u}$: generalized eigenvector Solve: $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{v}$

Degenerate Node

$\begin{cases} y_1' = 2y_1 \\ y_2' = y_1 + 2y_2 \end{cases}$ $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$

- Find eigenvalues and eigenvectors
- Find a generalised eigenvector
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r = 2$ (repeated)

Eigenvector $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ for r

Generalised eigenvector $\mathbf{u}$

Solve: $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{v}$

$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Get solution $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 1 \\ s \end{pmatrix}$ s is a parameter

Choose s = 0 for a specific solution $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Fundamental set of solutions

$\mathbf{x}_1(t) = e^{2t}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\mathbf{x}_2(t) = te^{2t}\mathbf{v} + e^{2t}\mathbf{u} = te^{2t}\begin{pmatrix} 0 \\ 1 \end{pmatrix} + e^{2t}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^{2t}\begin{pmatrix} 1 \\ t \end{pmatrix}$

General solution

$\mathbf{y} = c_1e^{2t}\begin{pmatrix} 0 \\ 1 \end{pmatrix} + c_2e^{2t}\begin{pmatrix} 1 \\ t \end{pmatrix}$

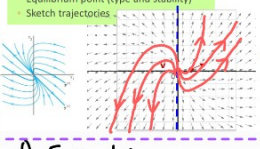
You can choose other values of s to get other generalized eigenvectors $\mathbf{u}$

- Find eigenvalues and eigenvectors
- Find a generalised eigenvector
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r = 2$ (repeated)

Eigenvector $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ for r

1 positive eigenvalue, only 1 indep. eigenvector
→ equilibrium point: degenerate node, unstable



Trajectories on opposite sides of the eigenvector move in opposite directions

Orientation of trajectories

- Test a point, say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Multiply by $\mathbf{A}$ to get $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$
- Tangent vector $\mathbf{y}'$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, pointing upward

0 Eigenvalue

$\begin{cases} y_1' = 2y_1 + y_2 \\ y_2' = 2y_1 + y_2 \end{cases}$ $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium points (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 0$ and $r_2 = 3$

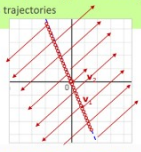
Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ for r_1, and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for r_2

Fundamental set of solutions

$\mathbf{x}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ $\mathbf{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}e^{3t}$

General Solution

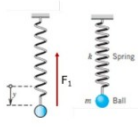
$\mathbf{y} = c_1\begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2\begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix}$



Mass Spring System

This is another classical example of modeling the free motions of a mass on a spring, using a 2nd order ODE.

We take an ordinary coil spring that resists extension as well as compression. Suspend it vertically from a fixed support and attach an iron ball at its lower end.



We now pull the ball down by a distance y as shown. In other words, y is the displacement of the ball from rest, and we take the downward direction as positive.

Now the spring will exert a spring force F_1. By Hooke's law, this force is proportional to the stretch y, with the spring constant k (> 0):

$$F_1 = -ky \quad \text{(i)}$$

The minus sign indicates that the force is against the displacement. It is a restoring force: It wants to restore the system, that is, to pull the ball back to y = 0.

Every system has damping. Otherwise it would keep moving forever. But if the damping is small and the motion of the system is considered over a relatively short time, we may disregard damping.

If we assume the system is undamped, then Newton's law gives the model:

$$my'' = F_1 \quad \text{(ii)}$$

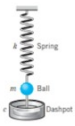
Combining (i) and (ii) gives this 2nd order ODE:

$$y'' = -\tfrac{k}{m}y$$

which describe a motion known as harmonic oscillation.

Damped Mass Spring System

We now connect the ball to a dashpot as shown to add a damping force F_2.



This damping force is given to be proportional to the velocity y', with the damping constant c.

$$F_2 = -cy' \quad \text{(iii)}$$

When the motion is downward, F_2 is an upward force; when the motion is upward, F_2 becomes a downward force. So the Newton's Second law gives

$$my'' = F_1 + F_2 = -ky - cy'$$

by using (i) and (iii).

So now we obtained the ODE of the damped mass–spring system:

$$my'' + cy' + ky = 0.$$

By substituting y and y' with y_1 and y_2, we can convert this ODE into a 2 x 2 SDE

$$\begin{matrix} y_1' = y_2 \\ y_2' = -\frac{k}{m}y_1 - \frac{c}{m}y_2 \end{matrix}$$

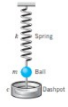
The matrix form of this SDE is given below:

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

4.4.16 Mass - Spring System

Recall that the mass-spring system is given by this 2nd order ODE: $my'' + cy' + ky = 0$

which can be converted into the SDE: $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$



Here k: Spring constant and c: damping constant. We may assume k and c > 0 except when there is no damping (c = 0).

We would like to find out the type of equilibrium point and its stability, under various conditions. In this context, the equilibrium point represents the stage when the ball is at rest.

The characteristic polynomial of the matrix $\begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}$ is given by:

$$\begin{vmatrix} \lambda & -1 \\ \frac{k}{m} & \lambda + \frac{c}{m} \end{vmatrix} = \lambda^2 + \frac{c}{m}\lambda + \frac{k}{m}$$

By solving for the roots of this quadratic polynomial, we get the eigenvalues r_1 and r_2 :

$$\frac{1}{2}\left(-\frac{c}{m} \pm \sqrt{\left(\frac{c}{m}\right)^2 - \frac{4k}{m}}\right).$$

Let's simply the expression by denoting

$$\Delta = \left(\frac{c}{m}\right)^2 - \frac{4k}{m}$$

So we have $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right)$ and $r_2 = \frac{1}{2}\left(-\frac{c}{m} - \sqrt{\Delta}\right)$.

We will consider a few cases:

Case 1: c = 0 (no damping)

In this case, $\frac{c}{m} = 0$ and $\Delta < 0$. So $r_1 = \frac{1}{2}\sqrt{\Delta}$ and $r_2 = -\frac{1}{2}\sqrt{\Delta}$ are purely imaginary complex eigenvalues. The equilibrium point is a center, which is neutrally stable.

Case 2: $c^2 > 4mk$

In this case, $\Delta > 0$. So $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right)$ and $r_2 = \frac{1}{2}\left(-\frac{c}{m} - \sqrt{\Delta}\right)$ are real distinct eigenvalues. Furthermore, both are negative numbers (see below *). The equilibrium point is a node and is asymptotically stable.

(*Clearly r_2 is negative. To see r_1 is also negative, note that $\Delta = \left(\frac{c}{m}\right)^2 - \frac{4k}{m} < \left(\frac{c}{m}\right)^2 \Rightarrow \sqrt{\Delta} < \frac{c}{m}$ by taking square root on both sides. Hence $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right) < 0$.)

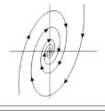
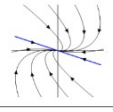
Case 3: $c^2 = 4mk$

In this case, $\Delta = 0$. So $r_1 = r_2 = -\frac{c}{2m}$ is a real repeated negative eigenvalue. The equilibrium point is a degenerate node and is asymptotically stable.

Case 4: $0 < c^2 < 4k$

In this case, $\frac{c}{m} > 0$ and $\Delta < 0$. So $r_1 = \frac{1}{2}\left(-\frac{c}{m} + \sqrt{\Delta}\right)$ and $r_2 = \frac{1}{2}\left(-\frac{c}{m} - \sqrt{\Delta}\right)$ are complex eigenvalues with negative real parts. The equilibrium point is a spiral and is asymptotically stable.

We summarize the 4 cases as follow:

Damping	Condition	Type of points	Trajectories
No damping (the ball will keep oscillating without coming to a halt)	c = 0	Centre	
Underdamping (the damping force is weak relative to the spring force; the ball will oscillate with smaller and smaller amplitude before coming to a stop.)	$c^2 < 4mk$	Spiral point	
Critical damping (the ball will come to rest as quickly as possible without oscillation)	$c^2 = 4mk$	Degenerate node	
Over damping (the damping force is strong relative to the spring force; the ball again will return to the equilibrium state without oscillating; the stronger the damping force, the slower will the ball come to rest)	$c^2 > 4mk$	Node	